

National Coverage Determination (NCD)

Lung Volume Reduction Surgery (Reduction Pneumoplasty)

240.1

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Tracking Information

Publication Number

100-3

Manual Section Number

240.1

Manual Section Title

Lung Volume Reduction Surgery (Reduction Pneumoplasty)

Version Number

2

Effective Date of this Version

01/01/2004

Ending Effective Date of this Version

11/17/2005

Implementation Date

01/05/2004

Description Information

Benefit Category

Inpatient Hospital Services

Outpatient Hospital Services Incident to a Physician's Service

Physicians' Services

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Item/Service Description

Lung volume reduction surgery (LVRS) or reduction pneumoplasty, also referred to as lung shaving or lung contouring, is performed on patients with severe emphysema in order to allow the remaining compressed lung to expand, and thus, improve respiratory function.

Indications and Limitations of Coverage

A - Covered Indications

Medicare-covered LVRS approaches are limited to bilateral excision of a damaged lung with stapling performed via median sternotomy or video-assisted thoracoscopic surgery.

1. National Emphysema Treatment Trial (NETT) participants (effective for services performed on or after August 11, 1997):
Medicare provides coverage to those beneficiaries who are participating in the NETT trial for all services integral to the study and for which the Medicare statute does not prohibit coverage.
2. Medicare will only consider LVRS reasonable and necessary when all of the following requirements are met (effective for services performed on or after January 1, 2004):
 - a. The patient satisfies all the criteria outlined below:

Assessment	Criteria
History and physical examination	Consistent with emphysema
	BMI, $\leq 31.1 \text{ kg/m}^2$ (men) or $\leq 32.3 \text{ kg/m}^2$ (women)
	Stable with $\leq 20 \text{ mg}$ prednisone (or equivalent) qd
Radiographic	High Resolution Computer Tomography (HRCT) scan evidence of bilateral emphysema
Pulmonary function (pre-rehabilitation)	Forced expiratory volume in one second (FEV ₁) $\leq 45\%$ predicted ($\geq 15\%$ predicted if age ≥ 70 years)
	Total lung capacity (TLC) $\geq 100\%$ predicted post-bronchodilator
	Residual volume (RV) $\geq 150\%$ predicted post-bronchodilator
Arterial blood gas level (pre-rehabilitation)	PCO ₂ , $\leq 60 \text{ mm Hg}$ (PCO ₂ , $\leq 55 \text{ mm Hg}$ if 1-mile above sea level)
	PO ₂ , $\geq 45 \text{ mm Hg}$ on room air (PO ₂ , $\geq 30 \text{ mm Hg}$ if 1-mile above sea level)
Cardiac assessment	Approval for surgery by cardiologist if any of the following are present: Unstable angina; left-ventricular ejection fraction (LVEF) cannot be estimated from the echocardiogram; LVEF $< 45\%$; dobutamine-radionuclide cardiac scan indicates coronary artery disease or

	ventricular dysfunction; arrhythmia (>5 premature ventricular contractions per minute; cardiac rhythm other than sinus; premature ventricular contractions on EKG at rest)
Surgical assessment	Approval for surgery by pulmonary physician, thoracic surgeon, and anesthesiologist post-rehabilitation
Exercise	Post-rehabilitation 6-min walk of ≥ 140 m; able to complete 3 min unloaded pedaling in exercise tolerance test (pre- and post-rehabilitation)
Consent	Signed consents for screening and rehabilitation
Smoking	Plasma cotinine level ≤ 13.7 ng/mL (or arterial carboxyhemoglobin $\leq 2.5\%$ if using nicotine products) Nonsmoking for 4 months prior to initial interview and throughout evaluation for surgery
Preoperative diagnostic and therapeutic program adherence	Must complete assessment for and program of preoperative services in preparation for surgery

b. In addition, the patient must have:

- Severe upper lobe predominant emphysema (as defined by radiologist assessment of upper lobe predominance on CT scan), or

c. Severe non-upper lobe emphysema with low exercise capacity.

Patients with low exercise capacity are those whose maximal exercise capacity is at or below 25 watts for women and 40 watts (w) for men after completion of the preoperative therapeutic program in preparation for LVRS. Exercise capacity is measured by incremental, maximal, symptom-limited exercise with a cycle ergometer utilizing 5 or 10 watt/minute ramp on 30% oxygen after 3 minutes of unloaded pedaling.

d. The surgery must be performed at facilities that were identified by the National Heart, Lung, and Blood Institute to meet the thresholds for participation in the NETT, and at sites that have been approved by Medicare as lung transplant facilities. These facilities are listed on our Web site at

www.cms.gov/coverage/lvrsfacility.pdf . The CMS is currently working to develop accreditation standards for facilities to perform LVRS and when implemented, will consider LVRS to be reasonable and necessary only at accredited facilities.

e. The surgery must be preceded and followed by a program of diagnostic and therapeutic services consistent with those provided in the NETT and designed to maximize the patient's potential to successfully undergo and recover from surgery. The program must include a 6- to 10-week series of at least 16, and no more than 20, preoperative sessions, each lasting a minimum of 2 hours. It must also include at least 6, and no more than 10, postoperative sessions, each lasting a minimum of

2 hours, within 8 to 9 weeks of the LVRS. This program must be consistent with the care plan developed by the treating physician following performance of a comprehensive evaluation of the patient's medical, psychosocial and nutritional needs, be consistent with the preoperative and postoperative services provided in the NETT, and arranged, monitored, and performed under the coordination of the facility where the surgery takes place.

B - Noncovered Indications

1. LVRS is not covered in any of the following clinical circumstances:

- a. Patient characteristics carry a high risk for perioperative morbidity and/or mortality;
- b. The disease is unsuitable for LVRS;
- c. Medical conditions or other circumstances make it likely that the patient will be unable to complete the preoperative and postoperative pulmonary diagnostic and therapeutic program required for surgery;
- d. The patient presents with FEV₁ ≤20% of predicted value, and either homogeneous distribution of emphysema on CT scan, or carbon monoxide diffusing capacity of ≤20% of predicted value (high-risk group identified October 2001 by the NETT); or
- e. The patient satisfies the criteria outlined above in section 2(a), and has severe, non-upper lobe emphysema with high exercise capacity. High exercise capacity is defined as a maximal workload at the completion of the preoperative diagnostic and therapeutic program that is above 25 w for women and 40 w for men (under the measurement conditions for cycle ergometry specified above).

2. All other indications for LVRS not otherwise specified remain noncovered.

Claims Processing Instructions

[TN 26 \(Medicare Claims Processing\)](#) 

[TN 27 \(Medicare Claims Processing\)](#) 

Transmittal Information

Transmittal Number

3

Coverage Transmittal Link

<https://www.cms.gov/Regulations-and-Guidance/Guidance/Transmittals/Downloads/R3NCD.pdf> 

Revision History

11/2003 - Expanded coverage to include patients who are: (1) Non high-risk and present with severe, upper-lobe emphysema; or, (2) Non high-risk and present with severe, non upper-lobe emphysema with low exercise capacity. Effective date 1/01/2004. Implementation date 1/5/2004. M+C Implementation date 4/5/2004. ([TN 3](#)) (CR 2688)

07/1997 - Provided coverage policy. Effective date 08/11/1997. (TN 102)

12/1995 - Provided noncoverage policy. Effective date NA. (TN 83)

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [First reconsideration for Lung Volume Reduction Surgery \(CAG-00115R\)](#)
- [Second reconsideration for Lung Volume Reduction Surgery \(CAG-00115R2\)](#)

Additional Information

Other Versions

Title	Version	Effective Between	
Lung Volume Reduction Surgery (Reduction Pneumoplasty)	3	11/17/2005 - N/A	View
Lung Volume Reduction Surgery (Reduction Pneumoplasty)	2	01/01/2004 - 11/17/2005	You are here
Lung Volume Reduction Surgery (Reduction Pneumoplasty)	1	08/11/1997 - 01/01/2004	View

